ONE RESIDUE CLASS MODULO EACH PRIME CAN COVER EVERY SUFFICIENTLY LARGE INTEGER

Abstract. We give an affirmative solution to Erdős Problem 279. For every fixed integer $k \geq 3$, one can choose a residue $0 \leq r_p < p$ for each prime p so that every sufficiently large integer n can be written as $n = r_p + tp$ for some prime p and integer $t \geq k$. The proof was generated by OpenAI's GPT-5.6-sol model. Its main ingredients are a large hub prime, a multiplicative semigroup, and a deterministic sieve that assigns fresh prime residue classes without destroying earlier coverage.

1. Introduction

Let $\mathbb{N} = \{1, 2, \ldots\}$ and let $\mathbb{P}$ denote the set of primes. For a fixed integer $k \geq 3$, a residue assignment is a sequence

$$\boldsymbol{r} = (r_p)_{p\in\mathbb{P}}, \qquad 0 \leq r_p < p.$$

It covers n at level k if

$$n = r_p + tp$$

for some $p \in \mathbb{P}$ and some $t \in \mathbb{Z}$, $t \geq k$. The problem considered here, also indexed as Erdős problem 279, asks whether for every $k \geq 3$ some assignment covers a complete tail of $\mathbb{N}$.

Theorem 1.1. *For every integer $k \geq 3$, there are a residue assignment $\boldsymbol{r} = (r_p)_{p\in\mathbb{P}}$ and an integer N such that every $n \geq N$ has a representation*

$$n = r_p + tp$$

with p prime and $t \geq k$.

The use of canonical representatives is part of the assertion.

Lemma 1.2 (Maturity equivalence). *If $0 \leq r_p < p$, then*

$$n = r_p + tp \text{ for some } t \in \mathbb{Z},\ t \geq k \quad \Longleftrightarrow \quad n \equiv r_p \pmod p \text{ and } kp \leq n.$$

Proof. Only the reverse implication needs comment. Congruence makes $t = (n - r_p)/p$ an integer. If $t \leq k - 1$, then

$$n = r_p + tp \leq (p-1) + (k-1)p = kp - 1,$$

contrary to $kp \leq n$. □

If $k_1 \leq k_2$, a cover at level k_2 is a cover at level k_1. Thus a construction for $k = 3$ alone would not prove Theorem 1.1; our construction treats an arbitrary fixed k.

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1.1. **Outline.** We shift the target by writing $m = n - 1$ and select shifted classes

$$a_p \equiv r_p - 1 \pmod{p}.$$

Fix a large prime $h > k$, set

$$G = \{p \in \mathbb{P} : p \equiv 1 \pmod{h}\}, \qquad \delta = \frac{1}{h-1},$$

and leave the shifted zero class on every prime outside $\{h\} \cup G$. The hub class $a_h = 1$ covers every G-smooth m not divisible by h. Apart from isolated prime targets, the only difficult integers are

$$m = h^e u, \qquad e \geq 1, \qquad u \in \langle G \rangle,$$

where $\langle G \rangle$ is the multiplicative semigroup generated by G.

The primes in G are partitioned into a finite hub set H, a periodic pool T used to cover isolated prime targets, and a large pool S used for multiscale cleanup. For

$$\rho = \sqrt{h/k}, \qquad X_{j+1} = \rho X_j,$$

the controller annuli are

$$R_j = S \cap \left(\frac{X_{j+1}}{h}, \frac{X_j}{k} \right].$$

They are disjoint and adjacent. A prime $p \in R_j$ is both mature for every target $m \in (X_j, X_{j+1}]$ and strictly greater than m/h.

The analytic core is a deterministic estimate, uniform in arbitrary previously selected nonzero classes on the small G-primes. It bounds the number of hard survivors at scale X by

$$\big(C_{\mathrm{sv}} + o(1)\big)\delta A_\delta v^\delta (\rho - 1)\frac{X}{\log X}, \qquad A_\delta = \frac{\mathrm{e}^{-\gamma\delta}}{\Gamma(\delta)},$$

where C_{sv} is absolute and v is fixed. Since

$$A_\delta < \delta, \qquad \frac{\rho - 1}{1/k - \rho/h} = \sqrt{kh},$$

this is smaller than the controller reservoir when h is large.

Permanence is supplied by a causal backup lemma. Changing a controller $p \in G$ from the provisional class 0 to a nonzero class can only remove a zero-class witness cp below the current frontier with $c < h$. When $c = 1$, the hub h is a backup; when $c > 1$, a prime divisor of c is a permanent outside zero-class backup.

1.2. **Analytic inputs.** We use Fouvry and Tenenbaum's Bombieri–Vinogradov theorem for multiplicative functions [2, Definition 1.1 and Theorem 1.8], the fixed-function Selberg–Delange estimate of de la Bretèche and Tenenbaum [1, Theorem 2.1], and the fundamental lemma for upper beta-sieve weights [3, Lemma 6.11], in the convenient formulation [4, (2.5)–(2.6), Theorem 2.2]. All hypotheses and constant dependencies are checked explicitly.

2. A deterministic sieve for the hard semigroup

Throughout this section, $h \geq 5$ is a fixed prime,

$$G = \{p \in \mathbb{P} : p \equiv 1 \pmod h\}, \qquad \delta = \frac{1}{h-1},$$

and

$$f(u) = \mathbf{1}_{u \in \langle G \rangle}.$$

The function f is completely multiplicative and takes values in $\{0, 1\}$.

Proposition 2.1 (Deterministic old-prime sieve)**.** *Fix $\rho > 1$. There is an absolute constant C_{sv} with the following property. Choose a sufficiently large fixed integer v. For every X, assign an arbitrary nonzero class a_p* mod p *to each*

$$p \in G, \qquad p \leq z := X^{1/v}.$$

Uniformly in all these choices, the number of integers

$$m = h^e u \in (X, \rho X], \qquad e \geq 1, \qquad u \in \langle G \rangle,$$

which avoid every assigned class is at most

$$\big(C_{\mathrm{sv}} + o_{h,\rho,v}(1)\big) \delta A_\delta v^\delta (\rho - 1) \frac{X}{\log X}, \qquad A_\delta = \frac{\mathrm{e}^{-\gamma\delta}}{\Gamma(\delta)}. \tag{2.1}$$

2.1. **Translated progression estimates.** Fix $e \geq 1$. For $p \in G$, define the nonzero class

$$b_{p,e} \equiv a_p h^{-e} \pmod p. \tag{2.2}$$

For a squarefree product d of primes in G, CRT gives a reduced class $b_{d,e}$ mod d satisfying the local congruences. Since $(d, h) = 1$, choose an integer representative with

$$\widetilde{b}_{d,e} \equiv b_{d,e} \pmod d, \qquad \widetilde{b}_{d,e} \equiv 1 \pmod h. \tag{2.3}$$

Then $(\widetilde{b}_{d,e}, dh) = 1$, while the progression modulo d is unchanged.

For $I = (Y, \rho Y]$, write

$$F(I) = \sum_{u \in I} f(u), \qquad F_d(I) = \sum_{\substack{u \in I \\ (u,d)=1}} f(u), \qquad A_d(I, b) = \sum_{\substack{u \in I \\ u \equiv b \ (\mathrm{mod}\ d)}} f(u).$$

The class $\mathcal{F}(D, K)$ in [2, Definition 1.1] consists of multiplicative functions g for which:

(i) there are $2 = \Upsilon_1 < \Upsilon_2 < \cdots$ satisfying
$$\frac{\Upsilon_{j+1}}{\Upsilon_j} \geq 1 + \frac{1}{(\log(2\Upsilon_j))^K};$$

(ii) $g(p) = g(p')$ whenever p, p' lie in the same corresponding interval and $p \equiv p'$ (mod D);

(iii) $|g(n)| \leq \tau_K(n)$.

Our f lies in $\mathcal{F}(h, 1)$: its value on primes depends only on the class modulo h, and $|f| \leq 1 = \tau_1$.

For every $A > 0$, [2, Theorem 1.8] supplies $B = B(A, 1)$ such that

$$\sum_{q \leq \sqrt{x}/(\log(3x))^B} \max_{(b,qh)=1} |\Delta_f(x; q, h, b)| \ll_{h,A} \frac{x}{(\log(3x))^A}. \tag{2.4}$$

When $(q,h)=1$, the discrepancy is

$$\Delta_f(x;q,b)=\sum_{\substack{u\le x\\ u\equiv b \pmod q}} f(u)-\frac{1}{\varphi(q)}\sum_{\substack{u\le x\\ (u,q)=1}} f(u).$$

Define

$$D_Y=\frac{\sqrt{Y}}{(\log(3\rho Y))^B}. \tag{2.5}$$

This is within the range in (2.4) at both $x=Y$ and $x=\rho Y$. Applying (2.4) at the endpoints, using (2.3), and restricting to squarefree G-supported moduli gives

$$\sum_{d\le D_Y}\max_{(b,dh)=1}\left|A_d(I,b)-\frac{F_d(I)}{\varphi(d)}\right|\ll_{h,A,\rho}\frac{Y}{(\log Y)^A}. \tag{2.6}$$

The maximum makes this uniform in every correlated vector of old residue classes.

2.2. **A uniform coprime mean by exact Möbius inversion.** The prime number theorem in the fixed progression 1 mod h gives

$$\sum_{p\le x} f(p)\log p=\delta x+O_h(x/\log^2 x).$$

For any fixed $1/2<\sigma_0<1$,

$$\sum_p\left\{\frac{f(p)^2}{p^{2\sigma_0}}+\sum_{\nu\ge 2}\frac{f(p^\nu)}{p^{\nu\sigma_0}}\right\}<\infty.$$

Thus [1, Theorem 2.1] applies with $A=2$ and $r=\delta$. In that theorem $\delta_{1,A}$ is Kronecker's delta, so $\delta_{1,2}=0$. Hence

$$M(x):=\sum_{u\le x} f(u)=\frac{K_h}{\Gamma(\delta)}x(\log x)^{\delta-1}\{1+O_h(1/\log x)\}, \tag{2.7}$$

where

$$K_h=\lim_{z\to\infty}K_h(z)>0,$$

$$K_h(z)=\prod_{p\le z}(1-1/p)^\delta\prod_{\substack{p\le z\\ p\in G}}(1-1/p)^{-1}. \tag{2.8}$$

No parameter-uniform contour theorem is required. If d is squarefree and supported on G, exact Möbius inversion gives

$$\begin{aligned} M_d(x)&:=\sum_{\substack{u\le x\\ (u,d)=1}} f(u)\\ &=\sum_{r\mid d}\mu(r)\sum_{\substack{u\le x\\ r\mid u}} f(u)=\sum_{r\mid d}\mu(r)M(x/r). \end{aligned} \tag{2.9}$$

The final identity holds because $r\in\langle G\rangle$ and $f(rv)=f(v)$.

Lemma 2.2 (Local density)**.** *Uniformly for squarefree $d\le\sqrt{Y}$ supported on G,*

$$\frac{F_d((Y,\rho Y])}{\varphi(d)}=\frac{F((Y,\rho Y])}{d}\left\{1+O_{h,\rho}\left(\frac{(\log\log(3Y))^3}{\log Y}\right)\right\}. \tag{2.10}$$

Proof. For $x \in \{Y, \rho Y\}$ and $r \mid d$, one has $x/r \ge \sqrt{Y}$. Inserting (2.7) into (2.9), using

$$\left(1 - \frac{\log r}{\log x}\right)^{\delta - 1} = 1 + O_h\left(\frac{\log r}{\log x}\right) \qquad (r \le \sqrt{x}),$$

and then applying

$$\sum_{r \mid d} \frac{1}{r} = \prod_{p \mid d}(1 + 1/p), \qquad \sum_{r \mid d} \frac{\log r}{r} = \prod_{p \mid d}(1 + 1/p) \sum_{p \mid d} \frac{\log p}{p + 1},$$

gives

$$M_d(x) = \frac{\varphi(d)}{d} M(x) \left\{1 + O_h\left(\frac{R(d)}{\log x}\right)\right\}, \tag{2.11}$$

where

$$R(d) = \prod_{p \mid d} \frac{p+1}{p-1} \left(1 + \sum_{p \mid d} \frac{\log p}{p+1}\right).$$

The maximal-order estimate $d/\varphi(d) \ll \log\log(3d)$ gives

$$\prod_{p \mid d} \frac{p+1}{p-1} \le \left(\frac{d}{\varphi(d)}\right)^2 \ll (\log\log(3Y))^2.$$

Also

$$1 + \sum_{p \mid d} \frac{\log p}{p+1} \ll \log\log(3Y).$$

Indeed, split at $Z = \log(3d)$; the contribution of $p \le Z$ is $O(\log Z)$, and the contribution of $p > Z$ is at most $Z^{-1} \sum_{p \mid d} \log p = O(1)$. Thus $R(d) \ll (\log\log(3Y))^3$.

Subtract (2.11) at Y and at ρY. Because $\rho > 1$ is fixed,

$$M(\rho Y) - M(Y) \asymp_{h,\rho} Y (\log Y)^{\delta - 1},$$

so both endpoint errors have the asserted relative size. □

2.3. **Translated upper sieve.** Let

$$P(z) = \prod_{\substack{p \le z \\ p \in G}} p.$$

Use upper beta-sieve weights λ_d^+ of level D_Y. Their required properties are

$$|\lambda_d^+| \le 1, \qquad \lambda_d^+ = 0 \text{ unless } d \mid P(z),\ d \text{ squarefree},\ d \le D_Y, \tag{2.12}$$

and

$$\mathbf{1}_{L=1} \le \sum_{d \mid L} \lambda_d^+ \qquad (L \mid P(z)). \tag{2.13}$$

For $g(d) = 1/d$, the sieve has dimension at most one, since

$$\prod_{\substack{w \le p < z \\ p \in G}} (1 - 1/p)^{-1} \le \prod_{w \le p < z} (1 - 1/p)^{-1} \le \frac{\log z}{\log w}\{1 + O(1/\log w)\}. \tag{2.14}$$

The constant is absolute. The fundamental lemma [3, Lemma 6.11] therefore supplies absolute constants s_0 and C_{sv} such that, if $D_Y \geq z^{s_0}$,

$$\sum_{d \mid P(z)} \frac{\lambda_d^+}{d} \leq C_{\mathrm{sv}} \prod_{\substack{p \leq z \\ p \in G}} (1 - 1/p). \tag{2.15}$$

More precisely, one may fix a beta-sieve parameter $\beta \geq 5$; for $s = \log D_Y / \log z \geq \beta + 1$, the left side equals the product times $1 + O(s^{-s/2})$ [4, Theorem 2.2].

For $u \in I$, define

$$L_e(u) = \prod_{\substack{p \leq z,\ p \in G \\ u \equiv b_{p,e} \pmod p}} p.$$

Survival is exactly $L_e(u) = 1$, and $d \mid L_e(u)$ is exactly the CRT fibre $u \equiv b_{d,e} \pmod d$. Thus (2.13) gives

$$\begin{aligned} S_e(I) &:= \sum_{\substack{u \in I \\ u \not\equiv b_{p,e} \pmod p \ (p \leq z,\ p \in G)}} f(u) \\ &\leq \sum_{d \mid P(z)} \lambda_d^+ A_d(I, b_{d,e}). \end{aligned} \tag{2.16}$$

Set

$$V_h(z) = \prod_{\substack{p \leq z \\ p \in G}} (1 - 1/p), \qquad \eta_Y = \frac{(\log\log(3Y))^3}{\log Y}.$$

Combining (2.6), (2.10), and (2.16) gives

$$S_e(I) \leq F(I) \sum_d \frac{\lambda_d^+}{d} + O_{h,\rho}\left(F(I) \eta_Y \sum_d \frac{|\lambda_d^+|}{d} \right) + O_{h,A,\rho}\left(\frac{Y}{(\log Y)^A} \right). \tag{2.17}$$

Now

$$\sum_d \frac{|\lambda_d^+|}{d} \leq \prod_{\substack{p \leq z \\ p \in G}} (1 + 1/p) \ll_h (\log z)^{\delta},$$

whereas $V_h(z) \asymp_h (\log z)^{-\delta}$. Relative to $F(I)V_h(z)$, the middle error in (2.17) is

$$\ll_h \frac{(\log\log(3Y))^3}{\log Y} (\log z)^{2\delta} = o(1),$$

because $h \geq 5$ gives $2\delta < 1$. Taking $A > 2$ makes the final error negligible. Hence

$$S_e(I) \leq \{C_{\mathrm{sv}} + o_{h,\rho}(1)\} F(I) V_h(z). \tag{2.18}$$

2.4. **Exact constant and summation over the layers.** The constant K_h in (2.7) also determines the fixed-progression Mertens formula. Indeed, (2.8) gives

$$V_h(z)^{-1} = K_h(z) \prod_{p \leq z} (1 - 1/p)^{-\delta} \sim K_h \mathrm{e}^{\gamma\delta} (\log z)^{\delta}$$

by the ordinary Mertens product. Equivalently,

$$V_h(z) \sim \mathrm{e}^{-\gamma\delta} K_h^{-1} (\log z)^{-\delta}. \tag{2.19}$$

Therefore

$$
\begin{aligned}
F((Y,\rho Y])V_h(z) &\sim \frac{\mathrm{e}^{-\gamma\delta}}{\Gamma(\delta)}(\rho-1)\frac{Y}{\log Y}\left(\frac{\log Y}{\log z}\right)^{\delta} \\
&= A_\delta(\rho-1)\frac{Y}{\log Y}\left(\frac{\log Y}{\log z}\right)^{\delta}. \qquad (2.20)
\end{aligned}
$$

Thus the h-dependent constant K_h cancels exactly.

Take $z = X^{1/v}$. For $m = h^e u \in (X,\rho X]$, put $Y_e = X/h^e$. If $Y_e \geq \sqrt{X}$, then

$$
\frac{\log D_{Y_e}}{\log z} = \frac{\frac{1}{2}\log Y_e - B\log\log(3\rho Y_e)}{(1/v)\log X} \geq \frac{v}{4} - o(1). \tag{2.21}
$$

Choose v so that $v/4 > s_0$. Since $(\log Y_e/\log z)^\delta \leq v^\delta$, equations (2.18) and (2.20) give

$$
S_e \leq \{C_{\mathrm{sv}} + o(1)\}A_\delta v^\delta(\rho-1)\frac{X/h^e}{\log(X/h^e)}. \tag{2.22}
$$

All error terms used above are uniform for $X^{1/2} \leq Y_e \leq X$. To justify the next asymptotic, first fix E and let $X \to \infty$ in the terms $e \leq E$. For the remaining terms, $\log(X/h^e) \geq \frac{1}{2}\log X$, so after division by $X/\log X$ their contribution is at most $2\sum_{e>E} h^{-e}$. Now let $E \to \infty$. Thus

$$
\sum_{\substack{e\geq 1 \\ X/h^e \geq \sqrt{X}}} \frac{X/h^e}{\log(X/h^e)} \sim \frac{X}{\log X}\sum_{e\geq 1} h^{-e} = \delta\frac{X}{\log X}. \tag{2.23}
$$

For the complementary layers the trivial estimate is

$$
\sum_{\substack{e: Y_e < \sqrt{X} \\ \rho Y_e \geq 1}} \big(|(Y_e,\rho Y_e]\cap\mathbb{N}| + 1\big) \ll_{h,\rho} \sqrt{X} + \log X = o(X/\log X). \tag{2.24}
$$

All remaining intervals contain no positive integer. Equations (2.22)–(2.24) prove Proposition 2.1.

Lemma 2.3. *For $0 < \delta < 1$, one has $A_\delta < \delta$.*

Proof. The functional equation and Weierstrass product for the gamma function give

$$
\frac{A_\delta}{\delta} = \frac{\mathrm{e}^{-\gamma\delta}}{\Gamma(1+\delta)} = \prod_{n\geq 1}\left(1+\frac{\delta}{n}\right)\mathrm{e}^{-\delta/n} < 1,
$$

because $\log(1+x) < x$ for $x > 0$. □

3. Prime targets and the controller reservoir

Fix an arbitrary integer $k \geq 3$. Choose the absolute integer v required in Proposition 2.1. We next choose a prime $h > k$, sufficiently large that

$$
2C_{\mathrm{sv}}v^\delta\sqrt{kh}\,A_\delta < 1, \qquad \delta = (h-1)^{-1}. \tag{3.1}
$$

This is possible by Lemma 2.3, since the left side is at most

$$
\frac{2C_{\mathrm{sv}}v^\delta\sqrt{kh}}{h-1} \longrightarrow 0
$$

as $h \to \infty$ through the primes.

Set
$$\rho = \sqrt{h/k}, \qquad G = \{p : p \equiv 1 \pmod{h}\}.$$
In the shifted variable $m = n - 1$, begin with
$$a_h = 1, \qquad a_p = 0 \quad (p \notin \{h\} \cup G). \tag{3.2}$$
Every class subsequently assigned on G will be nonzero.

3.1. **Covering isolated prime targets.** Choose a finite set $H \subset G$, and set $a_\ell = 1$ for $\ell \in H$. A sufficiently large prime target q is already covered if
$$q \equiv 1 \pmod{h} \quad \text{or} \quad q \equiv 1 \pmod{\ell} \text{ for some } \ell \in H.$$
By the prime number theorem in the finitely many reduced CRT classes, the complementary primes have density
$$\Delta_H = \left(1 - \frac{1}{h-1}\right) \prod_{\ell \in H} \left(1 - \frac{1}{\ell - 1}\right). \tag{3.3}$$
The series $\sum_{\ell \in G} 1/\ell$ diverges, so Δ_H can be made arbitrarily small.

Enlarge H until a number τ can satisfy
$$k\Delta_H < \tau < h\Delta_H, \qquad \tau < \delta/2. \tag{3.4}$$
We realize τ as a periodic prime density. Choose a sufficiently large auxiliary prime L, coprime to $h \prod_{\ell \in H} \ell$. Each pair of reduced congruences
$$p \equiv 1 \pmod{h}, \qquad p \equiv c \pmod{L}$$
defines a prime set of density $1/((h-1)(L-1))$. Unions of such classes have density mesh $\delta/(L-1)$. Thus, for large L, a union $T_0 \subset G$ has density τ satisfying (3.4). Put
$$T = T_0 \setminus H, \qquad S = G \setminus (H \cup T), \qquad \sigma = \mathrm{dens}_{\mathbb{P}}(S) = \delta - \tau > \delta/2. \tag{3.5}$$

Let $q_1 < q_2 < \cdots$ enumerate the complementary primes in (3.3), and let $p_1 < p_2 < \cdots$ enumerate T. Inverting the fixed-progression prime-counting asymptotics gives
$$\frac{p_i}{q_i} \longrightarrow \frac{\Delta_H}{\tau} \in (1/h, 1/k). \tag{3.6}$$
For every sufficiently large i, define
$$a_{p_i} \equiv q_i \pmod{p_i}. \tag{3.7}$$
This class is nonzero and $kp_i < q_i < q_i + 1$. Assign $a_p = 1$ to the finitely many unused early primes of T, and absorb the finitely many early q_i into the final threshold. Thus every sufficiently large shifted prime target is covered.

3.2. **Disjoint controller annuli.** Choose a starting scale X_0, to be enlarged below, and define
$$X_j = X_0 \rho^j, \qquad R_j = S \cap \left(\frac{X_{j+1}}{h}, \frac{X_j}{k}\right] \quad (j \geq 0). \tag{3.8}$$
Since $\rho^2 = h/k$, one has $X_{j+2}/h = X_j/k$. The half-open annuli R_j are pairwise disjoint and tile a tail of S. The fixed-modulus prime number theorem gives
$$|R_j| \sim \sigma \left(\frac{1}{k} - \frac{\rho}{h}\right) \frac{X_j}{\log X_j}. \tag{3.9}$$

Assign $a_p = 1$ to each S-prime $p \le X_1/h$. At the beginning of stage j, every G-prime $p \le X_j^{1/v}$ has a permanent nonzero class. This is clear for $H \cup T$. For S, the initial segment and earlier annuli suffice once X_0 is large, because

$$hX_j^{1/v} < X_j.$$

Call

$$m = h^e u, \qquad e \ge 1, \qquad u \in \langle G \rangle \tag{3.10}$$

a hard target. Apply Proposition 2.1 to $(X_j, X_{j+1}]$, ignoring every more recent prime. The number of hard targets not covered by old primes is at most

$$\{C_{\mathrm{sv}} + o(1)\}\delta A_\delta v^\delta (\rho - 1)\frac{X_j}{\log X_j}. \tag{3.11}$$

The identity

$$\frac{\rho - 1}{1/k - \rho/h} = \sqrt{kh} \tag{3.12}$$

and $\sigma > \delta/2$ show that the ratio of the leading coefficient in (3.11) to that in (3.9) is at most

$$2C_{\mathrm{sv}} v^\delta \sqrt{kh}\, A_\delta < 1.$$

After fixing h, choose X_0 so large that this strict inequality beats all error terms at every stage, all preceding asymptotics are valid, and

$$X_1/h > kh. \tag{3.13}$$

At stage j, list the surviving hard targets and the primes of R_j increasingly. Pair each survivor with a distinct prime $p \in R_j$. If p is paired with m, put

$$a_p \equiv m \pmod p; \tag{3.14}$$

give every unused prime in R_j the class $a_p = 1$.

Lemma 3.1 (Controller validity)**.** *The class in* (3.14) *is nonzero, and p is mature at the original target $m + 1$.*

Proof. Since $p > X_{j+1}/h \ge m/h$, divisibility $p \mid m = h^e u$ would imply $m/p \ge h$, a contradiction. Hence $m \not\equiv 0 \pmod p$. Moreover,

$$p \le X_j/k < m/k < (m+1)/k.$$

□

3.3. **Causal permanence.** We temporarily regard an unprocessed S-prime as having class zero.

Lemma 3.2 (Causal backup)**.** *Let $p \in R_j$. Changing a_p from* 0 *to an arbitrary nonzero class cannot make any positive shifted integer $x \le X_{j+1}$ uncovered, provided all other assignments already used below that frontier are kept.*

Proof. A point that could lose p as a zero-class witness has the form $x = cp$. Since $p > X_{j+1}/h$, we have $c < h$.

If $c = 1$, then $p \equiv 1 \pmod h$, so the permanent hub class $a_h = 1$ covers $x = p$. Its original target $p + 1$ is mature because $kh < p < p + 1$, by (3.13).

If $c > 1$, choose a prime divisor $d \mid c$. Then $d < h$, hence $d \notin \{h\} \cup G$. The permanent class $a_d = 0$ covers $x = cp$, and it is mature because

$$kd < kh < p < cp + 1.$$

□

It follows inductively that every hard target covered at a stage remains covered forever. The initial segment and adjacent annuli assign one nonzero class to every prime in S; there is no reselection.

4. Completion of the cover

Proposition 4.1. *With the assignment constructed in Section 3, every sufficiently large shifted integer m is covered by a prime p satisfying*

$$kp \le m + 1.$$

Proof. There are four cases.

Case 1: m is $\{h\} \cup G$-smooth and $h \nmid m$. Every prime factor of m then lies in G, hence is 1 mod h. Thus $m \equiv 1 = a_h \pmod{h}$. For $m + 1 \ge kh$, the hub is mature.

Case 2: m is $\{h\} \cup G$-smooth and $h \mid m$. Then m has the unique hard form

$$m = h^e u, \qquad e \ge 1, \qquad u \in \langle G \rangle,$$

and was permanently covered in its block.

Case 3: m has a prime factor $q \notin \{h\} \cup G$ with $q \le m/k$. The permanent outside class $a_q = 0$ covers m, and $kq \le m < m + 1$.

Case 4: m is not $\{h\} \cup G$-smooth and every outside prime factor exceeds m/k. Assume $m > k^2$. Since m is not $\{h\} \cup G$-smooth, it has at least one outside prime factor. Two outside prime factors, counted with multiplicity, would have product $> m^2/k^2 > m$, which is impossible. Therefore there is exactly one outside prime factor q, occurring to the first power. The cofactor $c = m/q$ is $\{h\} \cup G$-smooth and satisfies $c < k$. Every nontrivial generator of this semigroup is at least $h > k$, so $c = 1$. Hence $m = q$ is prime, and it is covered by the prime-target construction.

These cases exhaust all $m > k^2$. The finitely many early prime-target exceptions and the fixed maturity thresholds are absorbed by the phrase "sufficiently large" in the proposition. □

Proof of Theorem 1.1. Let M_0 exceed X_0, k^2, all finitely many early prime-target exceptions, and every fixed maturity threshold. By Proposition 4.1, for every $m \ge M_0$ there is a prime p such that

$$m \equiv a_p \pmod{p}, \qquad kp \le m + 1.$$

For each prime p, define r_p to be the unique element of $\{0, \ldots, p-1\}$ satisfying

$$r_p \equiv a_p + 1 \pmod{p}.$$

Take $N = M_0 + 1$. If $n \ge N$, put $m = n - 1$; then

$$n \equiv r_p \pmod{p}, \qquad kp \le n.$$

Lemma 1.2 gives $n = r_p + tp$ for an integer $t \ge k$. The assignment is fixed globally and uses exactly one canonical class modulo every prime. □

5. A compactness reformulation

Although compactness is not needed in the construction, it clarifies the global quantifier. Fix k. For $N \le M$, let $F(k, N, M)$ be the finite constraint problem that chooses one residue modulo every prime $p \le M/k$ and covers each $n \in [N, M]$ using a prime $p \le n/k$.

Proposition 5.1. *The following are equivalent:*

(i) *one residue assignment covers every sufficiently large integer at level k;*

(ii) *there is a fixed N such that $F(k, N, M)$ is satisfiable for every $M \ge N$.*

Proof. Let

$$\Omega = \prod_{p \in \mathbb{P}} \{0, \dots, p-1\},$$

with finite discrete factors. It is compact. For each n, the set

$$K_n = \{\boldsymbol{r} \in \Omega : \exists p \le n/k,\ r_p \equiv n \pmod{p}\}$$

is clopen and depends on finitely many coordinates. A solution of $F(k, N, M)$ is equivalent, after arbitrary extension on unused coordinates, to a point of

$$L_M = \bigcap_{n=N}^{M} K_n.$$

If (ii) holds, the sets L_M are nonempty, closed, and nested. Compactness gives

$$\bigcap_{M \ge N} L_M = \bigcap_{n \ge N} K_n \ne \varnothing,$$

which is (i). The converse is immediate by restriction. $\square$

Remark 5.2. Negating Proposition 5.1 gives

$$\forall N\ \exists M \ge N : F(k, N, M) \text{ is unsatisfiable.}$$

A finite cover with a threshold that changes with M therefore does not imply a global assignment.

Declarations

Generative AI contribution and proof provenance. Substantive generative-AI use: this is a GPT-5.6-sol-generated proof. OpenAI's GPT-5.6-sol model, operating as Codex in a user-directed research session, generated the central construction, the deterministic semigroup sieve argument, the proof synthesis, and an initial manuscript draft. Separate GPT-5.6-sol instances then performed line-by-line logical, adversarial, and source-checking passes. These were internal AI-assisted quality-control passes, not human peer review and not proof-assistant formal verification. Defects identified in those passes were corrected in the present manuscript.

GPT-5.6-sol is not listed as an author because a generative AI system cannot assume scholarly accountability. By submitting this manuscript, the named human author or authors certify that they have independently checked every mathematical claim and bibliographic entry, approved the final manuscript, and accept full responsibility for its correctness, originality, citations, disclosure, and compliance with journal policy. Editors should apply the AI-use policy of the target journal.

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Human author responsibility. The corresponding author accepts responsibility for answering questions about the argument and for investigating and resolving any concern about accuracy or integrity.

Data, code, funding, and competing interests. No empirical data or finite computational result is used as a mathematical premise. Every mathematical dependency is either proved in the text or stated as a cited theorem with its hypotheses checked. No external funding was used in the preparation of this manuscript. The author declares no competing interests.

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References

[1] R. de la Bretèche and G. Tenenbaum, *Remarks on the Selberg–Delange method*, Acta Arithmetica **200** (2021), no. 4, 349–369.

[2] É. Fouvry and G. Tenenbaum, *Multiplicative functions in large arithmetic progressions and applications*, Transactions of the American Mathematical Society **375** (2022), no. 1, 245–299.

[3] J. Friedlander and H. Iwaniec, *Opera de Cribro*, American Mathematical Society Colloquium Publications, vol. 57, American Mathematical Society, Providence, RI, 2010.

[4] P. Kurlberg, S. Lester, and L. Rosenzweig, *Superscars for arithmetic point scatters II*, Forum of Mathematics, Sigma **11** (2023), e42.